

# Method hierarchy

## Capacity-Balance MCS

Baseline

Classical hourly capacity check · no storage

## Naive Storage MCS

[FAIL] Biased high

Naive peak-shaving heuristic · proactive discharge

## SOC-Floor Storage MCS

[FAIL] Partially corrected

Reserve-aware heuristic · SOC floor on proactive P

## Emergency-Only Storage MCS

[PASS] Matches EUE

Surplus charging · discharge only at shortfall

## Event-Window Storage MCS

[PASS] Matches EUE & LOLH

LP solved around screened scarcity windows

## Full-Year ED

LP reference

Full-year economic dispatch LP · Gurobi · benchmark

## PCM-UCED

PCM validation

Full-year unit commitment MILP · validation only

Increasing computational cost